

Appendix: Supplementary information

Code files

File	Function
Code for running simulations	
00_load-packages.R	Loads packages.
00_process_worldpop.R	Computes population-weighted density (pwd_adm2.qs).
01_prepare_inputs.R	Ranks municipalities and selects top 100 for <i>Wolbachia</i> release. Calculates R_0 and transmission rates for every municipality.
02_run_batch.R	Runs model for one municipality. Compiles and equilibrates model. Runs four intervention scenarios with 500 particles (set in config.yaml). Script set up to run in parallel on HPC. Script run once for each municipality.
03_combine_batches.R	Combines individual batch outputs. Summarises the results.
03b_exclude_batches.R	Removes outlier/failed municipalities and recomputes the same outputs as 03_combine_batches.R. Note – only one municipality removed (IDN.9.7_1) due to a large negative number of cases averted in both scenarios involving wMel.
calculate-R0_fn.R	Code to derive R_0 from FOI values. Used in 01_prepare_inputs.R. Sourced from Cattarino et al. (2020)
process-demography.R	Produces age-stratified demographic datasets by municipality and year. Provided by Emilie Finch.
config.yaml	Sets model parameters.
denv-model.R	Transmission model written in odin2 which compiles to C++.
equilibrate_sim.R	Defines a function to run 150-year equilibration period for one municipality using one particle. Used in 02_run_batch.R.
scenario_sim.R	Defines a function to run simulations for one scenario for one municipality.
utils.R	Defines a set of functions used in pre-processing, post-processing and tracking failed runs.
run.R	Master script to test code locally. Runs batches sequentially.
Code for figure generation	
equilibration_analysis.R	Produces plots of the equilibration period.
wol_selection_heatmaps.R	Produces heatmaps of the metrics used in <i>Wolbachia</i> targeting.
burden_maps.R	Produces heatmaps of cumulative cases and hospitalisations per 10,000 population at the municipality level.
report_figures.R	Produces non-map figures.
wol_selection_figures.R	Produces figures illustrating municipality selection for <i>Wolbachia</i> release.
tables.R	Produces tables summarising the 15-year cumulative cases and hospitalisations.